

| Question | Answer                                                                                         | Marks     |
|----------|------------------------------------------------------------------------------------------------|-----------|
| 1(a)     | force per unit mass                                                                            | <b>B1</b> |
| 1(b)(i)  | $g = GM/r^2$<br>$= (6.67 \times 10^{-11} \times 1.0 \times 10^{13}) / (3.6 \times 10^3)^2$     | <b>C1</b> |
|          | $= 5.1 \times 10^{-5} \text{ N kg}^{-1}$                                                       | <b>A1</b> |
| 1(b)(ii) | mass = $(960 / 9.81) \text{ kg}$<br>weight on comet = $(960 / 9.81) \times 5.1 \times 10^{-5}$ | <b>C1</b> |
|          | $= 5.0 \times 10^{-3} \text{ N}$                                                               | <b>A1</b> |
| 1(c)     | similarity: e.g. both attractive/pointed towards the comet<br>e.g. same order of magnitude     | <b>B1</b> |
|          | difference: e.g. radial/non-radial<br>e.g. same (over surface)/varies (over surface)           | <b>B1</b> |

9702/42

## Cambridge International AS/A Level – Mark Scheme

May/June 2017

**PUBLISHED**

| Question | Answer                      | Marks |
|----------|-----------------------------|-------|
| 2(c)(i)  | mean/average speed/velocity | B1    |